

Scalability & Future Plan

Date	15 March 2024
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application is designed primarily for local deployment and supports only a limited number of concurrent users.	Reduced performance when many users access the system simultaneously.	High
2	The prediction model is based on Linear Regression, which may not capture complex relationships between health and lifestyle factors.	Prediction accuracy may decrease for diverse or large datasets.	High
3	The application currently stores limited user information and prediction history.	Difficult to perform long-term user analysis and trend monitoring.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local Flask server.	Deploy the application on cloud platforms such as Render, AWS, or Azure with load balancing to support a larger user base.
2	Data Storage	Uses CSV files and/or SQLite for storing data.	Migrate to MySQL or PostgreSQL to efficiently manage large datasets and user records.
3	Performance	Predictions are generated using a single Linear Regression model.	Optimize preprocessing, implement caching, and evaluate advanced machine learning models for faster and more accurate predictions.
4	Security		

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve the user interface and dashboard with advanced charts and responsive design.	1–2 Months	Better user experience and easier interpretation of prediction results.
Phase 2	Integrate advanced machine learning algorithms such as Random Forest, XGBoost, or Neural Networks to improve prediction accuracy.	2–4 Months	Higher prediction accuracy and better performance on diverse datasets.
Phase 3	Develop a mobile application and integrate cloud-based database storage with real-time synchronization.	4–6 Months	Increased accessibility and support for a larger number of users.

Phase 4	Add AI-powered personalized health recommendations, wearable device integration, and real-time health monitoring.	6–12 Months	Comprehensive well-being assessment, continuous monitoring, and personalized user guidance.
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